

REPORT – Data Engineering Contribution

Part A: Introduction

Mental health has become a critical concern in modern society, particularly among students who face increasing academic pressure, financial stress, and lifestyle challenges. Depression among students is a growing issue that can significantly impact academic performance, social relationships, and long-term well-being. Early identification of depression is essential to enable timely intervention and support.

This project aims to develop a **Student Depression Prediction System** using machine learning techniques. The system analyses various factors such as academic pressure, sleep patterns, financial stress, and lifestyle habits to predict whether a student is likely to experience depression. By leveraging data-driven approaches, the system provides a predictive tool that can assist educators, counsellors, and healthcare professionals in identifying at-risk individuals.

The importance of this project lies in its potential to contribute to proactive mental health management. Traditional methods of identifying depression often rely on self-reporting or clinical diagnosis, which may not always be timely or accessible. A machine learning-based system can serve as an early screening tool, complementing professional assessments.

Furthermore, this project demonstrates the integration of artificial intelligence into real-world applications by developing a fully functional web-based interface. This enhances usability and accessibility, allowing non-technical users to interact with the system effectively.

Part B: Technical Review (LO1)

The development of the Student Depression Prediction System required the exploration of multiple data science and machine learning techniques. This section reviews the key methods and tools used.

Data Preprocessing Techniques:- Data preprocessing is a crucial step in machine learning, as raw data often contains missing values, inconsistencies, and categorical variables. The following techniques were applied:

- **Handling Missing Values:**
Numerical features were imputed using the median, while categorical features were filled using the mode. This approach ensures minimal distortion of data distribution.
- **Encoding Categorical Variables:**
Label encoding was used to convert categorical variables (e.g., gender, dietary habits) into numerical form, making them suitable for machine learning algorithms.
- **Feature Scaling:**
Standardisation using a Standard Scaler was applied to ensure that features with different scales do not disproportionately influence the model.

Machine Learning Models

Two primary models were explored:

1. Logistic Regression
 2. Random Forest
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Part C: Design & Implementation (LO2)

System Overview:

The system consists of the components such as Data preprocessing pipeline, Machine learning model Web-based user interface.

Data Pipeline:- Raw Data → Cleaning → Encoding → Scaling → Model Training → Prediction

Model Implementation:- The Random Forest classifier was implemented using the Scikit-learn library. The dataset was split into training and testing sets (80/20 split), and the model was trained on the processed data.

System Architecture:

The system architecture integrates machine learning with web development:

User Input → streamlit user interface → Data Processing → Model Prediction → Output Display

Individual Contribution (Key Highlight):-

My primary contribution to this project was the **development of the streamlit application and system integration**. This included:

- Designing the user interface for input collection
- Connecting the frontend with the backend model
- Ensuring correct data formatting and preprocessing during prediction
- Displaying results dynamically to the user

The interface allows users to input various attributes such as age, academic pressure, CGPA, sleep duration, and financial stress. Upon submission, the system processes the input and provides a prediction indicating whether the student is likely to be depressed.

The screenshot provided demonstrates a fully functional interface where predictions are displayed clearly, enhancing usability.

Part D: Technical & Critical Evaluation (LO3, LO4)

Technical Evaluation

The model was evaluated using multiple performance metrics. The Random Forest classifier achieved strong results:

- **Accuracy:** High overall correctness (~85–90%)
- **Precision:** Reliable identification of depressed cases
- **Recall:** Effective detection of actual depressed students
- **F1-score:** Balanced performance between precision and recall

The confusion matrix indicated that the model performs well in distinguishing between depressed and non-depressed students, although some misclassifications remain.

The system also successfully integrates real-time prediction through the Flask interface, demonstrating practical applicability.

Critical and Social Evaluation

While the technical performance is strong, several ethical and social considerations must be addressed.

1. Bias in Data

The dataset may not represent all demographics equally. For example:

- Gender imbalance
- Cultural or regional limitations

Such biases can affect model fairness and lead to unequal predictions.

2. Risk of Misclassification

- **False Positives:** May cause unnecessary concern
- **False Negatives:** May fail to identify at-risk students

In mental health contexts, these errors can have serious consequences.

3. Privacy and Data Security

The system handles sensitive personal information. It is essential to ensure:

- Secure data storage
- Confidentiality of user inputs

4. Ethical Use of AI

The system should not replace professional diagnosis. Instead, it should:

- Act as a **support tool**
 - Assist decision-making
 - Encourage further evaluation by professionals
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Part E: Project Proposal & Management

Project Management Reflection

The project was successfully completed through structured planning and task allocation. Key achievements include:

- Development of a functional machine learning model
- Implementation of a user-friendly web interface
- Integration of backend and frontend components

Challenges Faced

- Data preprocessing inconsistencies
- Aligning user input with model format
- Debugging integration issues

These challenges were addressed through iterative testing and refinement.

Future Work Proposal

The project can be extended in several ways:

1. Model Improvement

- Use advanced algorithms (e.g., XGBoost, Neural Networks)
- Perform hyperparameter tuning

2. Data Expansion

- Collect larger and more diverse datasets
- Include additional psychological and behavioural factors

3. Deployment

- Deploy on cloud platforms (e.g., AWS, Heroku)
- Enable real-time access for institutions

4. Explainability

- Implement explainable AI techniques (e.g., SHAP)
- Provide reasons for predictions

5. Security Enhancements

- Add authentication systems
 - Ensure encrypted data handling
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Part F: Conclusion

This project successfully demonstrates the application of machine learning techniques in predicting student depression. The Random Forest model provided strong predictive performance, while the Flask application ensured usability and accessibility. The project highlights the potential of AI in supporting mental health initiatives, while also emphasising the importance of ethical considerations and responsible deployment.

Overall, this work contributes to the growing field of AI-driven healthcare solutions and provides a foundation for further research and development in mental health prediction systems.

SHU AI Transparency Scale (AITS)

AITS: 2

Descriptor: AI for Shaping

Transparency Statement: AI has been used to shape the initial and/or final parts of the activity.

AI Contributions: AI is used for shaping parts of the activity. This includes initial outlining, concept development, prompting thinking, and/or improving structure/quality of the final output.

Human Contribution: Most of the activity is human developed/generated. AI ideas and suggestions are refined and reviewed. AI outputs are used for discrete and specific goals/outcomes.